

PARTH SHARMA

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SUMMARY

Machine Learning Engineer (B.Tech, Artificial Intelligence & Data Science) with hands-on experience in **Python, machine learning, deep learning, SQL, and model deployment**. Builds data-cleaning, feature-engineering, and ETL pipelines on 500K+ records across banking, aviation, and academic domains, and trains and compares classification models (Logistic Regression, Decision Trees, Gradient Boosting) for predictive analytics. Strong SQL and data-validation foundation for production-ready ML workflows.

EXPERIENCE

Machine Learning Engineer Intern — Decode Data Academy July 2025 – Present

- Built and maintained **ETL pipelines** (MySQL, SQL) that ingest, clean, and standardize 10,000+ records across 5+ branches and 2 colleges, eliminating 95%+ of data errors and creating reliable, **training-ready datasets**.
- Designed normalized relational databases, 10–15 SQL views, and 4–5 star-schema tables for scalable, high-performance data access.
- Made recurring workflows repeatable by integrating a **Pentaho Spoon** ETL layer into existing SQL reporting, improving efficiency by 20–30% and reducing manual maintenance.
- Added data validation, QA, and business-rule verification across 1,000+ records, essential for data quality in ML workflows.
- Delivered pipeline-backed reporting through 6 Power BI dashboards, refreshed weekly for end users.

PROJECTS

Insurance Incentive Optimization ([GitHub](#), [Live](#)) Python, FastAPI, Scikit-learn

- Built a renewal-probability classification pipeline on 67,875 policy records: median imputation, one-hot/target encoding, scaling, and an engineered `Total_late` feature in one **scikit-learn Pipeline**; compared Random Forest and XGBoost.
- **Productionized** the trained pipeline (joblib/pickle) behind a FastAPI `/predict` endpoint with Pydantic validation, CORS, and batch inference, deployed on Render.
- Layered a revenue-optimization step on the model's probabilities to recommend the best agent incentive (Rs. 500–4,500) per policy; web frontend live on Vercel.

Loan Default Risk Prediction ([GitHub](#)) Python, Pandas, Scikit-learn

- Built an end-to-end **ML pipeline** on a 55K-row, 122-column loan dataset: missing-value and outlier handling, encoding, and model-ready data preparation.
- Engineered **10+ predictive features** (income segmentation, credit ratio, EMI burden, employment duration) to improve customer risk profiling.
- Trained and compared **Logistic Regression, Decision Tree, and Gradient Boosting** classifiers to predict loan default risk; evaluated with standard classification metrics.

- Integrated **7 heterogeneous ride datasets** (465K+ raw records) into one schema; removed rate-card rows and 7K+ duplicates to produce 376K model-ready rows.
- Engineered cyclic (sin/cos) time features, location-frequency encodings for ~12.8K locations, and flagged/imputed distances; **detected and removed target-leaking columns** (e.g., weather, ratings).
- Built a leakage-free pipeline (train-only scalers, 70/15/15 split); trained HistGradientBoosting baselines and **PyTorch embedding MLPs** with early stopping and a 3-seed ensemble.

EDUCATION

- 2022 - 2026 Bachelor of Technology (Artificial Intelligence and Data Science) at **Lakshmi Narain College of Technology, Bhopal, Madhya Pradesh, India**
- 2020 - 2021 Higher Secondary Education (Class XII) at **Manikala Nahar Higher Secondary School**
- 2018 - 2019 Secondary Education (Class X) at **Shri Maniklal Nahar International School**

TECHNICAL SKILLS

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| Programming: | Python, SQL |
| Machine Learning: | Predictive Modeling, Feature Engineering, Exploratory Data Analysis, Classification (Logistic Regression, Decision Trees, Gradient Boosting), Model Evaluation |
| Deep Learning: | Deep Learning, Neural Networks |
| Deployment: | Model Deployment, FastAPI, REST APIs |
| Libraries: | Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Matplotlib, Seaborn |
| Data Engineering: | ETL Pipelines, Pentaho Spoon, Data Validation, Star Schema |
| Databases: | MySQL, Relational Databases |
| Tools: | Git, Jupyter Notebook, Microsoft Excel |
| Coursework: | Machine Learning, DBMS, Statistics, Data Structures and Algorithms |

CERTIFICATIONS

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| – AWS Academy Data Engineering Badge | Dec 2025 |
| – Wipro TalentNext Database Solutions Expert | Oct 2024 |
| – HackerRank SQL – 5-Star (Gold) Badge | |